CHARLES DARWIN UNIVERSITY
DARWIN CAMPUS

PRT661 DATA SCIENCE PRACTICE

ASSESSMENT 1 – PROJECT PROPOSAL AND DESIGN

HOUSING AFFORDABILITY FORECASTING SYSTEM FOR GREATER DARWIN REGION

Theme 2 – Predictive Analytics and Forecasting

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Project overview and objectives

The housing market has seen significant transformation in the Greater Darwin region due to several fluctuations in housing prices. Beyond the immediate effects on individuals, the Northern Territory Government has noted the widespread social problems associated with housing affordability, such as homelessness and overcrowding (Northern Territory Government, 2020).

The proposed Housing Affordability Forecasting System will model the affordability of a range of suburbs of Greater Darwin based on previous historical property, demographic and socioeconomic data. The system will integrate information on property sales with ABS income, population, rent and household data, as well as geographic properties like distance to Darwin CBD. The project will create predictive models to measure trends in affordability.

The main objectives are to:

- Gather property sales, demographic and socioeconomic data for the Greater Darwin Region
- Clean up the data and create features like number of bedrooms, distance from the CBD, and so on.
- Make and compare regression models to predict the value of a house.
- Create indicators of affordability based on relationships between property prices, household income, rents and mortgage costs.
- Name the factors that affect housing affordability.
- Create a dashboard showing historical data, forecasts and affordability data by suburbs.

Theme Justification

The project is directly relevant to Theme 2 – Predictive Analytics and Forecasting as historical data will be used to forecast housing market and affordability. The primary predictive task is regression in which property prices are considered a continuous variable.

It is not a primary concern of the project that real-time streaming data is being transmitted, therefore Theme 1 is not suitable. Theme 3 is not the central theme either as it doesn't concentrate on detecting unusual events or anomalies.

Problem Statement

The Greater Darwin Region has high housing affordability issues (NTCOSS, 2024). Household incomes and rental conditions vary between suburbs, and there could be significant variations in property prices between suburbs. Existing housing information only offers limited suburb-level affordability analysis.

The proposed system aims to fill this gap by predicting property prices and affordability metrics. The system will provide users with feedback on where affordability pressures are growing as well as what factors are driving the housing costs. First-home buyers, renters, real-estate organisations, investors and government planners are examples of potential stakeholders.

The system doesn’t guarantee the future price of the properties but is devised to serve as a decision supporting tool.

Initial Architecture Diagram

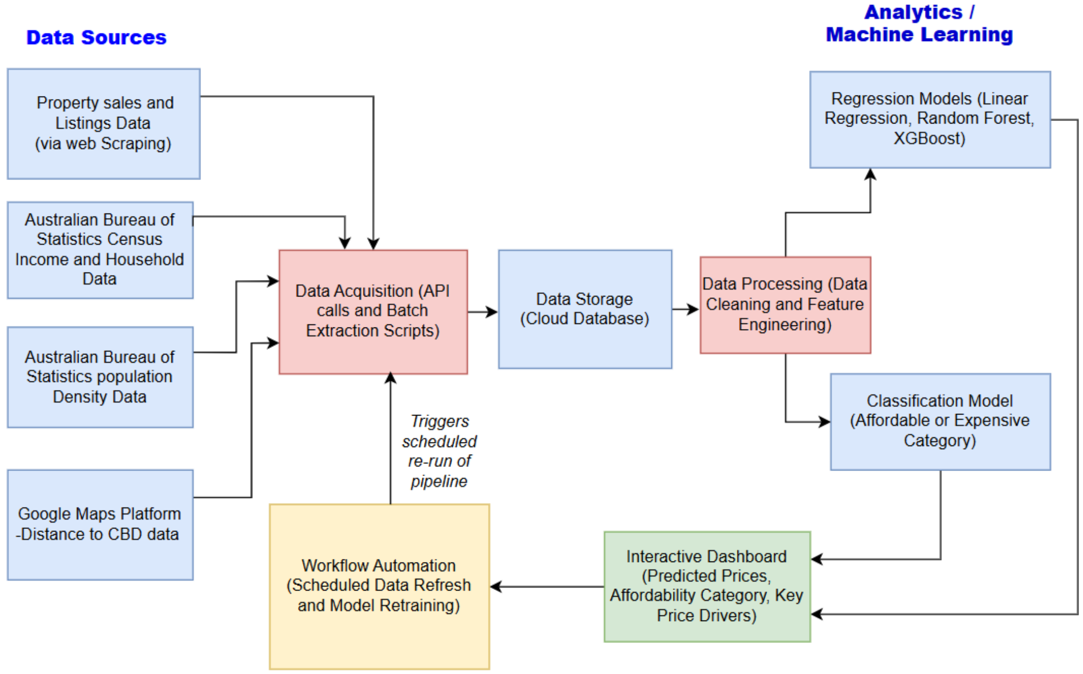

Figure 1: Initial Architecture Diagram

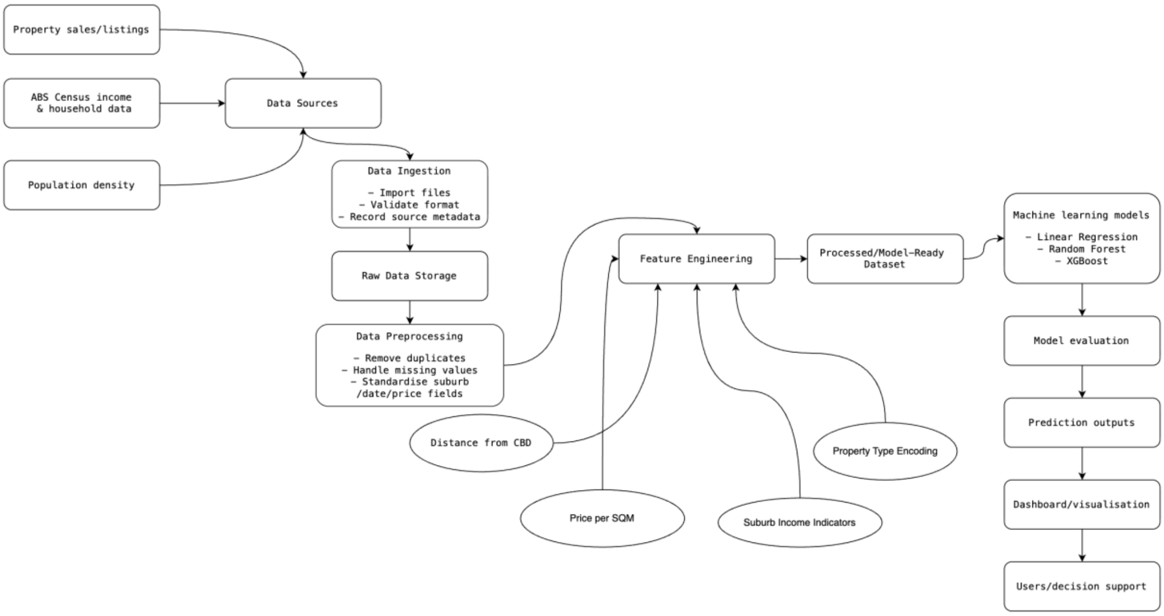

Figure 2: Component Interaction Diagram

The data will be gathered from available public sources (Homely, n.d.). ABS Census will be used for demographic and socioeconomic data. Geographic information will be captured using Google Maps API for computing location-based features like distance from Darwin CBD.

Workflow Plan

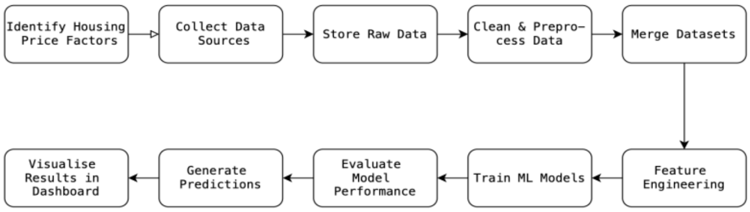

Figure 3: Workflow Plan

Data, Storage and Analytics Approach

About 15,423 sold property records scrapped from homely.com will be used in the project, and this will be supplemented by ABS Census and regional population data.

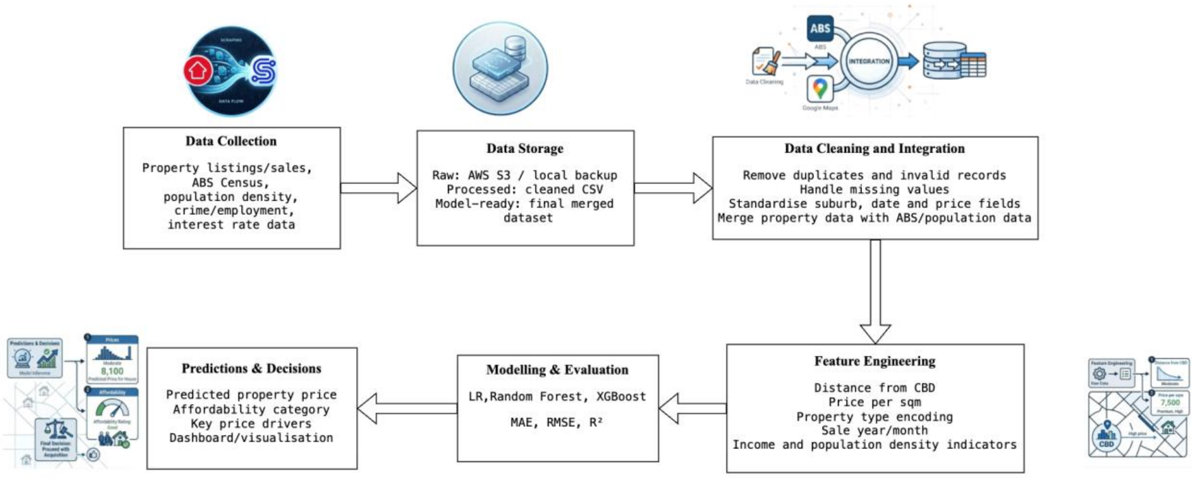

Figure 4: Data Pipeline Diagram

Data will be organised into four layers:

- Raw data: Datasets which have been downloaded or collected.
- Processed data: Cleaned and standardised datasets.
- Model-ready data: Data with engineered features built into the dataset.
- Output data: Predictions, evaluation metrics and feature-importance results.

Duplicate records will be removed, missing values will be dealt with, prices will be validated and examination of outlying data will be carried out. Property values can be estimated in the absence of values based on the medians of property prices.

Three regression models will be used: Linear Regression, Random Forest and XGBoost. Linear Regression will be used to establish a baseline model. Random Forest and XGBoost will be tested for modeling nonlinear relationships. The most successful model will be chosen based on model evaluation evidence. RMSE, MAE and R^2 will be used as evaluation measures. As the project is forecasting problem, temporal separation of training and testing data will be considered to minimize future data leakage.

Task Allocation and Team Structure

Team Member	Primary Role
Bipana Tripathee	Project Coordinator & Business Analyst
Alan Joshi John	Data Engineer
Roshan Neupane	System Architect & Machine Learning Developer
Lisa Vong	Project Planning & Documentation Coordinator

Aashish Sharma	Risk, Ethics & Visualisation Specialist

Table 1: Task Allocation

The team will use Jira for project management, GitHub for version control, and Microsoft teams for communication.

RACI Matrix

R = Responsible

A = Accountable

C = Consulted

I = Informed

Project Activity	**Bipana**	**Alan**	**Roshan**	**Lisa**	**Aashish**
Define project scope and objectives	I	A	C	R	I
Requirements analysis	I	A	C	R	I
System architecture design	R	C	A	I	I
Workflow and data pipeline design	I	R	A	C	I
Data acquisition	I	R	A	C	I
Data preprocessing & feature engineering	I	R	A	C	I
Machine learning model development	C	R	A	I	I
Model evaluation & validation	C	R	A	I	I
Dashboard visualisation development	I	C	A	C	R
Risk, ethics & privacy analysis	A	I	C	C	R
Documentation & report integration	C	C	I	A	R
Final presentation	R	R	R	R	R

Table 2: RACI Matrix

Initial Risk Analysis

S.N.	Risk	Likelihood	Impact	Mitigation
1.	Incomplete or noisy dataset	Medium	High	Use multiple datasets and apply data validation
2.	Model overfitting	Medium	Medium	Use train/test/validation splits and k-fold cross-validation before exploring advanced techniques
3.	Mismatch between data at different levels	Medium	Medium	Introducing a spatial aggregation and feature engineering step in the ETL pipeline
4.	Team-related delays or version conflicts	Low	Medium	Use Git branching workflows, weekly stand-ups, and task tracking.

Table 3: Initial Risk Analysis

Ethics, Privacy, Security Considerations

The project does not collect data that may identify any individual but instead makes use of statistical data gathered from the public domain. But the information that is available to the public will be dealt with carefully and all unnecessary personal information will not be stored.

Bias in the model, inequity between suburbs in data, and misinterpretation of forecasts are some of the ethical issues. Predictions will thus be made with proper uncertainty information and limitations. The APIs will not be opened in the public GitHub repository.

References

Homely. (n.d.). *Find suburb by region: Darwin, Northern Territory*. Retrieved August 16, 2026, from homely.com.au

Northern Territory Council of Social Service. (2024). *NTCOSS cost of living report: Issue 42 - The cost of housing in the Northern Territory*. https://ntcoss.org.au

Northern Territory Government. (2020). *A home for all Territorians: Northern Territory housing strategy 2020–2025*. Department of Territory Families, Housing and Communities. https://tfhc.nt.gov.au/__data/assets/pdf_file/0010/765433/nt-housing-strategy-2020-2025.pdf

Appendix

Jira Board: https://vongngocloi.atlassian.net/jira/software/projects/PDSPDT/boards/38/backlog
Github: https://github.com/alanjoshi/Data_Science_Practice_Grp10